# JANGHO KIM

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## RESEARCH PROFILE

Research in efficient and deployable AI, spanning model compression, knowledge distillation, quantization, pruning, and efficient inference for language and vision-language models. Applications include autonomous perception, image and video restoration, generative AI, personalization, and machine unlearning.

- **Recent publications:** five papers accepted at **NeurIPS 2026**; additional 2026 papers at CVPR, ECCV, KDD, DAC, AISTATS, ICDM, CIKM, IROS, and PETS/PoPETs.
- **Industry and government research:** principal investigator for projects with Hyundai Motor/Kia, LG Electronics CTO, POSCO, Easywith, ETRI, and NRF.

## PROFESSIONAL EXPERIENCE

**Kookmin University, Seoul** *Sep. 2022 - Present*
Assistant Professor, School of Artificial Intelligence
Machine Learning & Pattern Recognition (ML&PR) Laboratory

**NAVER WEBTOON AI Research Lab, Pangyo** *Jan. 2022 - Aug. 2022*
Research Scientist

**Qualcomm AI Research, Seoul** *Jul. 2020 - Jul. 2021*
Research Intern, Machine Learning

**Qualcomm AI Research, San Diego, CA** *Jul. 2019 - Oct. 2019*
Research Intern, Model Quantization

## EDUCATION

**Seoul National University, Korea** *Sep. 2017 - Feb. 2022*
Ph.D., Program in Intelligent Systems, Department of Transdisciplinary Studies
Advisor: Prof. Nojun Kwak

**Pohang University of Science and Technology (POSTECH), Korea** *Mar. 2015 - Feb. 2017*
M.S., Computer Science and Engineering
Advisor: Prof. Daijin Kim

**Dongguk University, Korea** *Mar. 2009 - Feb. 2015*
B.S., Information and Communication Engineering, *cum laude*

## SELECTED AWARDS AND HONORS

- **Selected Early-Career Researcher**, Korea Computer Congress (KCC) 2023
- **2nd place**, Machine Unlearning Challenge, hosted at NeurIPS 2023
- **1st place**, DCASE Challenge, Task 1A 2021
- Yulchon AI Scholarship (AI Star) 2021
- Graduate Sponsorship, Samsung Advanced Institute of Technology 2020 - 2021
- **3rd place**, ImageNet Classification, MicroNet Challenge at NeurIPS 2019

## SPONSORED RESEARCH AND INDUSTRY COLLABORATION

**Principal Investigator (PI)** *11 projects and advisory engagements*

**Easywith** *Mar. 2026 - Jun. 2026*
Research advisory on AI models for content IP protection and color transfer.

**Hyundai Motor & Kia, Robotics Lab** *Sep. 2025 - Sep. 2026*
AI model compression for real-time services using vision-language models.

*Related result:* Q-ViK [C2]: **78.6% lower KV-cache memory** and **13.6% lower decoding latency** than full cache (LLaVA-OneVision-Qwen2-7B; RTX 4090; 20% visual-KV retention).

**National Research Foundation of Korea (NRF) / MSIT** *Sep. 2025 - Aug. 2026*
A Hardware-Aware Model Compression Framework for Real-Time Execution of Next-Generation AI Models in Edge Devices.
*Young Researcher Seed Research; grant RS-2025-23524783.*

*Related result:* SharedKD [C12]: **4.1× inference speedup** over the teacher (3.70 to 0.91 ms) and **68% fewer parameters** (CenterPoint; NVIDIA DGX Spark, TensorRT FP16).

**POSCO** *Jul. 2025 - Feb. 2026*
Generation and detection of hard-to-distinguish irregular obstacles in steelworks.

**Easywith** *Jul. 2025 - Oct. 2025*
Research advisory on face-specialized AI models and video color transfer.

**Easywith** *May 2025 - Apr. 2026*
Generative AI for image/video analysis and generation (Year 2): high-resolution outpainting and video generation.

**LG Electronics, CTO** *Mar. 2025 - Dec. 2025*
Development of model quantization algorithms for on-device AI.

*Related result:* TAQ [C8]: **3.56× speedup over FP32** (2.987 to 0.839 s) with static INT8 BasicVSR on NVIDIA Jetson Orin Nano using TensorRT.

**Hyundai Motor & Kia, Autonomous Vanguard Team** *Nov. 2024 - Nov. 2025*
Exploratory research on road-icing recognition for safe Level-3 autonomous driving in low-temperature environments.

**Easywith** *Aug. 2024 - Feb. 2025*
Generative AI for image/video analysis and generation: high-resolution synthesis, outpainting, and image enhancement.

**Hyundai Motor & Kia, Robotics Lab (via Hyundai NGV)** *Apr. 2024 - Mar. 2025*
Development of model compression technology for vision transformers.

*Related result:* SERo [C15]: **2.4× higher throughput** than original DeiT-Base (401.4 to 960.3 images/s) and **68.8% fewer parameters** (NVIDIA A6000; batch size 64).

**Electronics and Telecommunications Research Institute (ETRI)** *Jul. 2023 - Nov. 2023*
Development of defect detection technology for environments with differing visual characteristics.

**PM** *1 program*

**Kookmin University, AI Star Fellowship Program** *Apr. 2025 - Dec. 2030*
Research and talent development in multimodal embodied AI and robotics; supported by MSIT and IITP.

**Participating Researcher** *2 projects*

**Korea Hydro & Nuclear Power (KHNP)** *Jul. 2026 - Jan. 2027*
Consulting on the development of a decision-making framework for dam operations.

**Rural Development Administration (RDA)** *Apr. 2024 - Dec. 2027*
Advancement of a farm-specific early warning system for agrometeorological hazards (Phase 2).

## SELECTED TALKS AND PRESENTATIONS

- **HMG TECH SUMMIT 2026, Hyundai Motor Group** October 6, 2026
  "Development of Model Compression Technology for Vision Transformers."
  Industry-academia research outcomes presentation #5; 16:30-17:00 (scheduled).
- **LG Electronics, CTO** May 23, 2025
  "Analysis and Applications of Self-Attention Architectures."
- **Suwon District Prosecutors' Office** March 31, 2025
  "Generative AI and DeepSeek."
- **Hyundai Motor, Robotics Lab** June 20, 2024
  Invited seminar.
- **Seoul St. Mary's Hospital, College of Medicine** 2021
  "Introduction to Deep Learning."
- **Korean Conference on Computer Vision (KCCV)** 2021
  Doctoral Colloquium: "Toward On-device Deep Learning."
- **NAVER Corp.** 2018
  Tech Talk: "Model Compression with Factor Transfer."

## PUBLICATIONS - 2026

*Jangho Kim is shown in bold; * denotes equal contribution where specified.*

[C1] Minkyu Kim, Baekseung Kim, Junhoo Lee, **Jangho Kim**, Nojun Kwak. "ManiFusion: Unlocking High-Throughput Generation via Superposition in Manifold Space." **NeurIPS 2026 (accepted)**.

[C2] Solhee Hwang, **Jangho Kim**. "Q-ViK: Question-Guided Visual KV Cache Eviction for Large Vision-Language Models." **NeurIPS 2026 (accepted)**.

[C3] Hea Eun Lee, Hyungi Lee, **Jangho Kim**. "PriSM: Prior-guided Shared-basis Mixture Personalization for LLMs under Sparse User Histories." **NeurIPS 2026 (accepted)**.

[C4] Jinwoo Chung, **Jangho Kim**. "Hardware-Friendly Token-Group Activation Quantization for Low-Bit Mamba Super-Resolution." **NeurIPS 2026 (accepted)**.

[C5] Hyungi Lee, **Jangho Kim**. "Improving Neural Processes in the Low-Data Regime via Context-Subset Training and Self-Distillation." **NeurIPS 2026 (accepted)**.

[C6] Hea Eun Lee, **Jangho Kim**. "Learning Task-Conditioned Distributional Initializers for Variational Quantum Algorithms." **ICDM 2026**.

[C7] Hyunjoon Cho, Jongyun Shin, Doun Lee, **Jangho Kim**. "Quantized Contrastive Unlearning." **CIKM 2026**.

[C8] Jinwoo Chung, Sangho An, Sungyeop Jung, **Jangho Kim**. "TAQ: Static-Deployable Temporal-Aware Quantization for Real-World Video Super-Resolution." **ECCV 2026**.

[C9] Jinwoo Chung, Sangho An, Hyunsung Park, **Jangho Kim**. "Sparse Recovery Bird's-Eye View Model Soup for Robust Camera-Based 3D Object Detection." **IROS 2026**.

[C10] Jinwoo Chung, Sungyeop Jung, Weronika Czorapinska, **Jangho Kim**. "Quantized Model Soup Shake-Up: Weight Perturbation for Enhanced Ensemble Diversity." **KDD 2026**.

[C11] Junhyoung Lee*, Seongwoon Jo*, Jeong-Hun Park, Yeonji Ryou, Jeongha Yang, **Jangho Kim**. "Non-linear Color Transfer via Learnable Bezier Flows." **CVPR 2026**.

[C12] Hyunjoon Cho*, Sangho An*, Hyunsung Park, **Jangho Kim**. "SharedKD: Gradient-Guided Dynamic Student Discovery inside a Single 3D Object Detector." **DAC 2026**.

[C13] HyeonJun Lee*, Hyeonsik Jo*, Jinwoo Chung, **Jangho Kim**. "SQuaT: Self-Supervised Knowledge Distillation via Student-Aware Quantized Teacher Features." **AISTATS 2026**.

## PUBLICATIONS - EARLIER CONFERENCES AND WORKSHOPS

[C14] Jinwoo Kim, Jongyun Shin, Sangho An, **Jangho Kim**. "Exploring Diverse Sparse Network Structures via Dynamic Pruning with Weight Alignment." **CIKM 2025**.

[C15] Sangho An, Jinwoo Kim, Keonho Lee, Jingang Huh, Chanwoong Kwak, Yujin Lee, Moonsub Jin, **Jangho Kim**. "Sparse Structure Exploration and Re-optimization for Vision Transformer." **UAI 2025**.

[C16] Jongyun Shin, Seunjin Han, **Jangho Kim**. "Cooperative Meta-Learning with Gradient Augmentation." **UAI 2024**.

[C17] **Jangho Kim**, Jayeon Yoo*, Yeji Song*, KiYoon Yoo, Nojun Kwak. "Finding Efficient Pruned Network via Refined Gradients for Pruned Weights." **ACM Multimedia 2023**.

[C18] Jiho Jang, KiYoon Yoo, Seonhoon Kim, Chaerin Kong, **Jangho Kim**, Nojun Kwak. "Self-Distilled Self-Supervised Representation Learning." **WACV 2023**.

[C19] Byeonggeun Kim, Seunghan Yang, **Jangho Kim**, Hyunsin Park, Juntae Lee, Simyung Chang. "Domain Generalization with Relaxed Instance Frequency-wise Normalization for Multi-device Acoustic Scene Classification." **Interspeech 2022**.

[C20] **Jangho Kim***, Jun-Tae Lee*, Simyung Chang, Nojun Kwak. "Variational On-the-Fly Personalization." **ICML 2022**.

[C21] KiYoon Yoo, **Jangho Kim**, Jiho Jang, Nojun Kwak. "Detection of Word Adversarial Examples in Text Classification: Benchmark and Baseline via Robust Density Estimation." **Findings of ACL 2022**.

[C22] Jeesoo Kim*, **Jangho Kim***, Jaeyoung Yoo, Daesik Kim, Nojun Kwak. "Vehicle Image Generation Going Well With the Surroundings." **ICONIP 2021**.

[C23] Byeonggeun Kim, Seunghan Yang, **Jangho Kim**, Simyung Chang. "Domain Generalization of Efficient Acoustic Scene Classification using Residual Normalization." **DCASE Workshop 2021**.

[C24] **Jangho Kim**, Simyung Chang, Nojun Kwak. "PQK: Model Compression via Pruning, Quantization, and Knowledge Distillation." **Interspeech 2021**.

[C25] **Jangho Kim**, Simyung Chang, Sungrack Yun, Nojun Kwak. "Prototype-based Personalized Pruning." **ICASSP 2021**.

[C26] **Jangho Kim**, KiYoon Yoo, Nojun Kwak. "Position-based Scaled Gradient for Model Quantization and Sparse Training." **NeurIPS 2020**.

[C27] **Jangho Kim**, Minsung Hyun, Inseop Chung, Nojun Kwak. "Feature Fusion for Online Mutual Knowledge Distillation." **ICPR 2020**.

[C28] Inseop Chung, SeongUk Park, **Jangho Kim**, Nojun Kwak. "Feature-map-level Online Adversarial Knowledge Distillation." **ICML 2020**.

[C29] **Jangho Kim***, Jeesoo Kim*, Nojun Kwak. "StackNet: Stacking Parameters for Continual Learning." **CVPR Workshop on Continual Learning in Computer Vision, 2020**.

[C30] **Jangho Kim**, SeoungUK Park, Nojun Kwak. "Paraphrasing Complex Network: Network Compression via Factor Transfer." **NeurIPS 2018**.

[C31] **Jangho Kim**, Yong-Joong Kim, Yonghyun Kim, Daijin Kim. "Detecting Korean Characters in Natural Scenes by Alphabet Detection and Agglomerative Character Construction." **IEEE SMC 2016**.

## JOURNAL ARTICLES

[J1] Doun Lee, Jongyun Shin, Jinwoo Bae, Hyunjoon Cho, **Jangho Kim**. "Weight Initialization based on Gradient Similarity for Versatile Machine Unlearning." **Proceedings on Privacy Enhancing Technologies (PoPETs), 2026; PETS 2026**.

[J2] Jongyun Shin*, Jinwoo Kim*, **Jangho Kim**. "Entropy-Guided Meta-Initialization Regularization for Few-Shot Text Classification." **Knowledge-Based Systems, 2025**.

[J3] Jihye Back, Namhyuk Ahn, **Jangho Kim**. "Magnitude Attention-Based Dynamic Pruning." **Expert Systems with Applications, 2025**.

[J4] **Jangho Kim**, JunHoo Lee, Donghoon Han, Nojun Kwak. "Layerwise-Priority-Based Gradient Adjustment for Few-Shot Learning." **Expert Systems with Applications, 2025**.

[J5] Sangho An, Jongyun Shin, **Jangho Kim**. "Quantization-Aware Training with Dynamic and Static Pruning." **IEEE Access, 2025**.

[J6] **Jangho Kim**. "Quantization Robust Pruning With Knowledge Distillation." **IEEE Access, 2023**.

[J7] **Jangho Kim**, KiYoon Yoo, Nojun Kwak. "Model Compression via Position-Based Scaled Gradient." **IEEE Access, 2022**.

## PREPRINTS AND TECHNICAL REPORTS

[T1] Byeonggeun Kim, Seunghan Yang, **Jangho Kim**, Simyung Chang. "QTI Submission to DCASE 2021: Residual Normalization for Device-Imbalanced Acoustic Scene Classification with Efficient Design." **DCASE Technical Report, 2021**.

[T2] **Jangho Kim***, Yash Bhalgat*, Jinwon Lee, Chirag Patel, Nojun Kwak. "QKD: Quantization-aware Knowledge Distillation." **arXiv, 2019**.